HAOPENG (JACK) XIE

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RESEARCH INTERESTS

- Computational Cognitive Science
- Natural Language Processing

EDUCATION

Johns Hopkins University **Sep 2023-May 2027**

Combined Bachelor's/Master's in Computer Science, Minor in Mathematics

- GPA: 3.81/4.0

PUBLICATIONS & MANUSCRIPTS

- **Xie, H.**, Ulgen, I. R., Zhao, S., Sisman, B., & Koehn, P. (2026). Lexical Tone and Word Order Matter in Translation: Prosodic Correspondence Across Languages. Submitted to NAACL. Project Page
- **Xie, H.**, Ulgen, I. R., Son, S., Sisman, B., & Koehn, P. (2026). Is Prosody Lost in Translation? Fine-Grained Cross-Lingual Prosody Similarity Across Languages. Findings of EMNLP 2026. Project Page
- Jin, C., Li, B., **Xie, H.**, Fang, C. M., Li, T., Longpre, S., Gu, H., Chen, M., & Shu, T. (2026). ThoughtTrace: Understanding User Thoughts in Real-World LLM Interactions. RLxF Workshop at ICML 2026 (Best Paper Award). Project Page

RESEARCH EXPERIENCE

Center for Language and Speech Processing (CLSP), JHU | Sep 2025 – Present

Researcher — Advisors: Prof. Philipp Koehn and Prof. Berrak Sisman

Prosody Analysis (2025–Present)

- Led a multilingual prosody study spanning 300K utterances; built the pipeline for speech filtering, bilingual alignment, acoustic feature extraction, and statistical analysis.
- Extended the analysis to lexical tone and word-order differences across 82K utterance pairs; found that removing tone-related pitch partly narrows cross-language correspondence gaps, while greater word reordering is associated with weaker local correspondence.
- First-author paper accepted to Findings of EMNLP 2026; follow-up manuscript under review at NAACL.

Prosody Transfer (2026–Present)

- Developing models to test whether source-language speech improves target-language prosody prediction beyond target-language text alone.
- Designing word- and sentence-level transfer experiments comparing acoustic features with representations of emphasis, hesitation, and confidence.

Social Cognitive AI (SCAI) Lab, JHU | Mar 2024 – Present
Researcher — Advisor: Prof. Tianmin Shu

ThoughtTrace (2025–2026)

- Contributed to the development of a benchmark for studying human thought processes during real-world LLM interactions; developed evaluation pipelines and conducted experiments comparing predicted thoughts with human annotations to assess how effectively user thoughts could be recovered from interaction context. Third-authored paper received Best Paper Award at RLxF@ICML 2026.

AI Reviewer (2026–Present)

- Lead app developer for a collaborative project studying AI-assisted scientific peer review; designed and implemented the experimental platform and integrated the AI reviewing agent.
- Built the infrastructure for recording AI conversations, reviewer actions, review outputs, timing logs, and experimental metadata, enabling downstream analysis of review quality and human–AI collaboration.

Multi-Agent Cooperation (2026–Present)

- Designing a text-based simulation to study how autonomous AI agents communicate, divide tasks, and coordinate toward shared objectives with limited information.
- Developing an evaluation framework comparing emergent cooperation, fixed-role teams, and centralized planning using task success, coordination failures, and communication cost.

HONORS & SERVICES

- **Best Paper Award**, ICML 2026 Workshop on RLxF
- **Reviewer**, ICML 2026 Workshop on Pluralistic Alignment
- **Dean's List**, Johns Hopkins University — 6 consecutive semesters
- **Ignite Fund Recipient**, Johns Hopkins University
- **USA Computing Olympiad (USACO), Platinum Division**

INTERNSHIP & WORKING EXPERIENCE

Supio, Software Engineer Intern **Jun 2025-Aug 2025**

- Developed an ML platform for a legal AI company and trained an intent classifier to improve automated processing of lawsuit files.

Teaching Assistant, Data Structures **Jan 2025-Jun 2025**

- Held office hours and graded homework assignments, programming work, and examinations.